Cantera vs NN | Test IC 45 | NRMSE [%]

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	1.39	6.72	37.3	18.6	1.34	66.6	6.28	4.55	17.9	N/A	Complete
2 P-only DeepONet10	**10k**	42.2	28.1	17	96.9	13	176	36.4	5.65	23	N/A	Complete
2 P-only DeepONet10	**long**	34.4	25.1	18.1	96.1	12.7	114	30.1	0.39	1.84	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	25	17.4	16.9	34.5	7.44	142	23.5	6.96	40	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	1.76	3	7.42	5.03	1.69	10.2	2.81	1.23	1.43	N/A	Complete
4 T/P DeepONet10	**10k**	23.1	18.8	34.8	14.5	18.6	74.6	19.9	11.3	16.7	N/A	Complete
4 T/P DeepONet10	**long**	60.6	3.47	105	11.7	3.11	384	33.4	16.4	14.1	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	11.6	2.98	18.1	16.9	3.36	70.5	6.49	3.36	2.12	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	51	7.85	58.3	14.2	3.68	391	33.4	11.8	18.7	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	14.1	13.5	16.1	26.9	13.9	62.9	13.4	8.09	17.3	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	40.1	3.43	64.4	8.96	1.87	266	23.3	8.51	9.3	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	10	3.21	18.5	16.8	4.53	70.5	4.99	2.52	1.92	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	1.18	0.546	1.56	0.834	0.464	9.47	0.898	0.383	0.923	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.